Hotel Management System NEA Analysis

Introduction:

The aim of this project is to create a hotel management system that assists staff with key hotel operations such as storing and managing guest information, room bookings, room availability, and payments.

As many hotels still use physical records and spreadsheets, this project is a more efficient way of managing records as it is more accessible to staff ,data is much less likely to get lost, duplicated information will not be needed and much less time consuming. This can be achieved by having an easy-to-use user interface and a secure network that connects all the staff devices.

The hotel is a non-inclusive resort. It has 4 blocks of hotels with 40 rooms in each block and 4 floors (10 rooms per floor). It has a swimming pool in the center with a swim up bar where guests can buy drinks. There are 3 restaurants with different cultures and they serve breakfast during the day.

The system is specifically intended for reception staff, managers and administrators and the system will have a hierachy so certain staff will have more access to information than others.

Problem identification

The hotel is managing a very large amount of information such as it has 4 blocks with 160 rooms in total. As guests increase a more accurate overall management will be needed but more mistakes are likely to happen. Any delays in updating bookings can cause bookings to be allocated incorrectly and worst case scenario a domino effect of overall inconsistent bookings. This can be solved by implementing a locking feature where as soon as a room is in the process of being booked the receptionist will click on the room and it will be greyed out and no other staff member can also select that room. Especially if they are about to book in a customer at the same time.

Another issue is security. The hotel will be holding personal information about each guest such as your full name, home address, email address, phone number, identification (passport or drivers license), date of birth, credit card details. It is the responsibility of the hotel to protect this data. This can be done by having a two step verification for each staff members login to ensure that no one who shouldn't have access should get into the system. Also encrypting the personal data can protect it if a data breach ever occured.

Stakeholder identification

Reception staff will use the system to:

- Check guests in and out
- Create, edit or cancel bookings
- View room availability
- Record guest details
- Process payments

Managers will use the system to:

- Monitor hotel performance
- Approve refunds or discounts
- View occupancy reports
- View financial reports
- Monitor booking statistics
- Manage room prices

Administrators will use the system to:

- Create and manage staff accounts
- Assign user permissions
- Update room type and prices
- Maintain the data base
- Configure system settings

Hotel guests:

- Benefit from accurate booking, quick service and secure handling of their personal information

Functional Requirement:

User login:

The system will allow staff sign into the system with a work email assigned to them by the administrators and their own password to go with it. All users accounts will have their role assigned to it, which can be reception, manager or administrator. This ensures that only authorized staff have access, and it also allows for permission levels depending on the users' responsibilities.

Room booking:

The system will allow staff to create, edit, and cancel bookings. It will show availability of the rooms and lock rooms that are in the process of being booked, so staff do not book the same room at the same time by accident. This will improve accuracy of room allocation and reduce errors.

Guest records:

The system will also store information about guests such as their full name, home address, email address, phone number, identification (passport or drivers license), date of birth,43 credit card details. Reception staff should be able to search for existing guests

Non functional requirements:

The system must require staff to log in using a work email, password and two factor authentication. Confidential data about guests should be encrypted and staff should only have access based on their role.

The system should load bookings, guest records and room availability within 2 seconds quickly. It should support the 160 rooms without slowing down.

The system should save changes immediately and create automatic daily backups. It shouldbe available all the time to reduce risk of data loss and make sure that bookings remain accurate.

The system should a have a clear and simple interface with labelled buttons and easy navigation. Error messages should explain any problems the system is facing. This allows staff to learn system quickly and reduce mistakesThe system should allow administrators to update room prices, room types and staff accounts easily without affecting existing data. The code and database should be organized so that future changes or bug fixes can be made.

Constraints:

This project will need to be finished in the window of time available of computer science aqa NEAs which means not every feature of an official hotel management system can be implemented but it will instead focus on the core features. Some users will have a different level of computer experience, but this can be prevented by quick training and an easy-to-

use interface. Another constraint is that the system must be able to run on the hotels existing hardware and should not require expensive hardware to run. Also the project will be developed using my current technical knowledge and resources meaning some features like cloud synchronisation cannot be implemented.

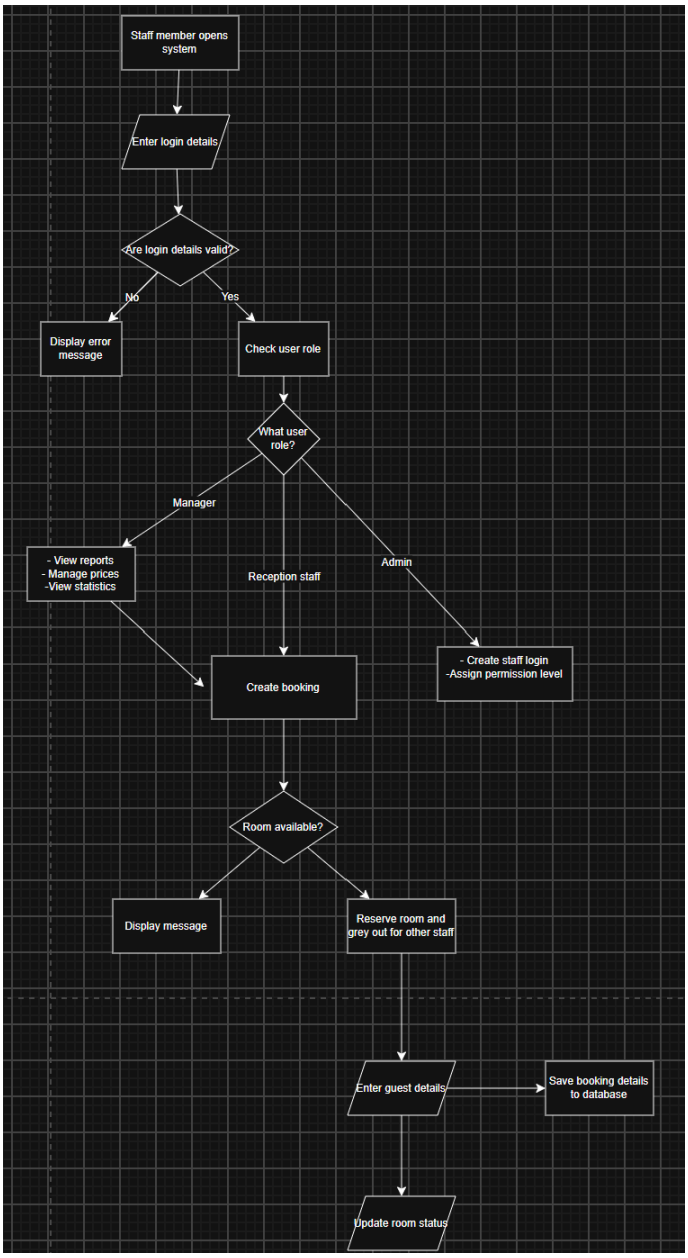
Success Criteria:

- ☐ Staff can log in using their work email and password
- ☐ Different staff have different permission levels
- ☐ Staff can create, edit and cancel bookings
- ☐ Double bookings of rooms is prevented
- ☐ Guest records can be added, searched, updated or deleted in a database
- ☐ Room availabiity updates automatically after bookings are changed
- ☐ Managers can view booking and occupancy reports
- ☐ Administrators can manage staff accounts and room information
- ☐ Bookings and guest records load quickly
- ☐ The system automatically saves changes across all devices
- ☐ The user interface is simple and easy for staff to use
- ☐ The system can run on low end computers

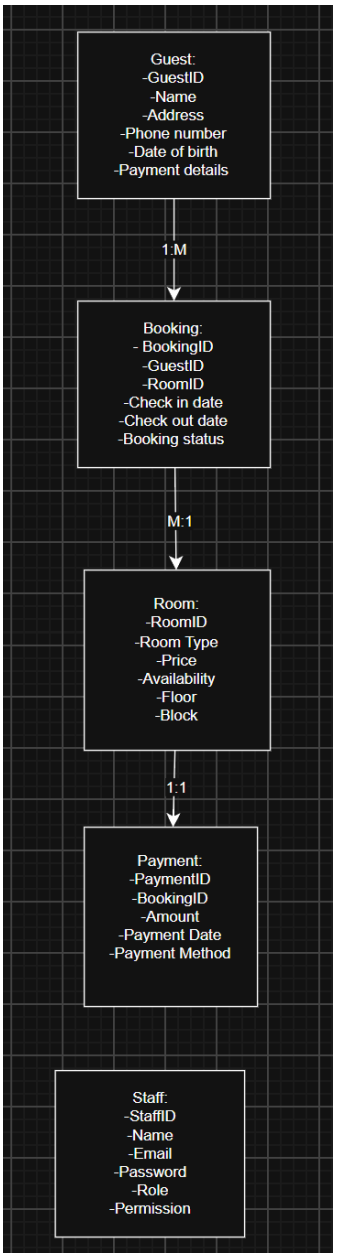
Modelling:

Case diagram:

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Entity relationship diagram:



Analysis Summary:

The analysis has identified that the hotel requires a computerised management system to efficiently manage guest information, room bookings, payments and staff access. Due to the size of the hotel an organised system is needed to reduce errors, improve efficiency and securely store data.

The system will provide reception staff, managers and administrators with different levels of access based on their responsibilities. Key features will include secure login, booking

management, guest records, payment processing reporting and staff permission management.

The requirements and models created have helped the main features, database structure and processes needed for the system. The next stage of the project will foces on designing the user interface, database, and algorithms required to implement the proposed solution.